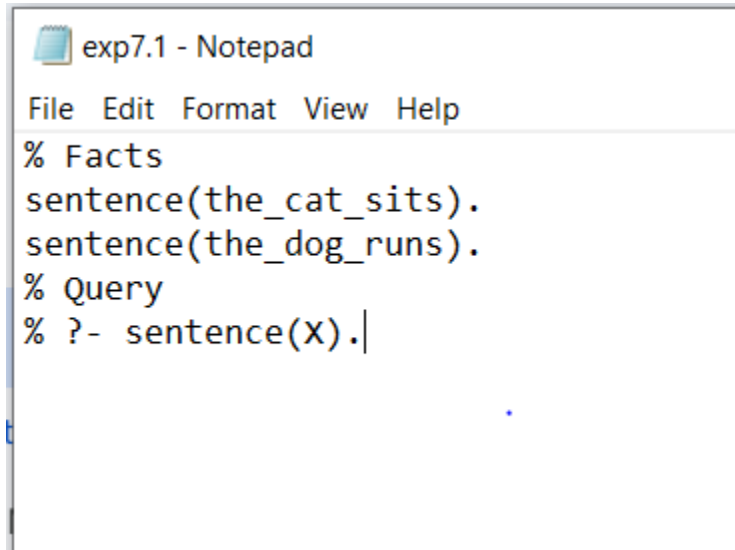


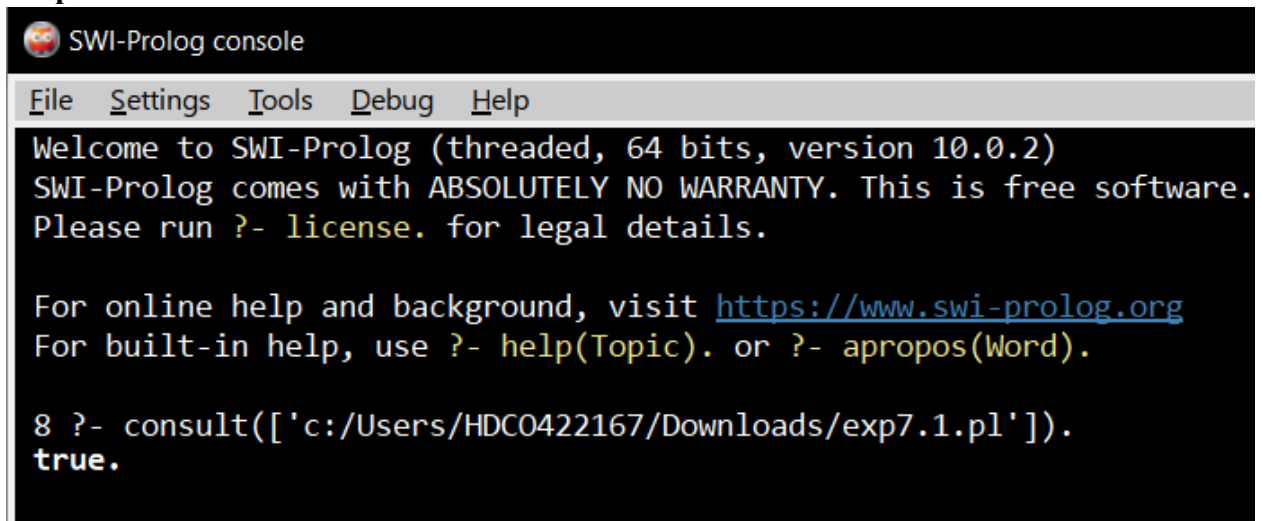
Unification and Resolution – Exercises with Solutions

Exercise 1: Sentence Matcher



```
exp7.1 - Notepad
File Edit Format View Help
% Facts
sentence(the_cat_sits).
sentence(the_dog_runs).
% Query
% ?- sentence(X).
```

Output:



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

8 ?- consult(['c:/Users/HDC0422167/Downloads/exp7.1.pl']).
true.
```

Exercise 2: Compound Term Unification



exp7.2 - Notepad

File Edit Format View Help

```
% Facts
likes(john, food(pizza, spicy)).
likes(mary, food(pizza, sweet)).
% Query
% ?- likes(john, food(pizza, T)).|
```

Output:

```
10 ?- likes(john, food(pizza, T)).
T = spicy ▲
```

Exercise 3: Conflict Resolution

 exp7.3 - Notepad
File Edit Format View Help
% Facts
rule1(climate_change).
rule2(climate_change).
% Rule
conflict(X) :- rule1(X), rule2(X).
% Query
% ?- conflict(climate_change).

Output:

```
12 ?- consult(['c:/Users/HDC0422167/Downloads/exp7.3.pl']).  
true.  
  
13 ?- conflict(climate_change).  
true.
```